

# comparative - analysis -



# How parameters affect on model

|                  | increase                                                                                                            | decrease                                |
|------------------|---------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| n_estimators     | better accuracy<br>more training time & memory usage                                                                | Fast training<br>Less stable prediction |
| max_depth        | Overfitting<br>Captures complex patterns                                                                            | Underfitting<br>Simple model            |
| min_samples_leaf | Reduce Overfitting<br>May underfit                                                                                  | Overfitting<br>More precise splits      |
| random_state     | This doesn't change performance, It makes the results repeatable. * Therefore always keep this fixed when comparing |                                         |
| n_jobs           | It will help to speed up training. There is no effect on accuracy.                                                  |                                         |

MAE - average absolute difference between actual vs predicted values

\*  $MAE \ll \text{target range [test]}$

\*  $\text{test\_MAE} \approx \text{Train\_MAE}$ , if  $\text{Test\_MAE} \gg \text{Train\_MAE}$  Overfitting

$R^2$  [coefficient of Determination] - How much variance is explained by the model

$R^2$

$> 0.9$  Excellent

$0.7 - 0.9$  Good

$0.5 - 0.7$  Acceptable

$0 - 0.5$  Weak

$< 0$  X

\*  $\text{Train } R^2 \uparrow$  but  $\text{test } R^2 \downarrow \rightarrow \text{Overfitting}$

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

MAPE

$< 5\%$  Excellent

$5\% - 10\%$  Very good

$10\% - 20\%$  Acceptable

$> 20\%$  Poor

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

$R^2 = 1$  ✓

$R^2 = 0$  Predict mean

$R^2 < 0$  X

$$MAPE = \frac{1}{n} \sum_{i=1}^n \frac{|y_i - \hat{y}_i|}{|y_i|} \times 100\%$$

target range = 0 - 800

epoch = 100

batch size = 32

Verbose = 0

### Test\_data

| Rev.center 1 | MAE  | R <sup>2</sup> | MAPE  | Rev.center 2 | MAE | R <sup>2</sup> | MAPE |
|--------------|------|----------------|-------|--------------|-----|----------------|------|
| Breakfast    | 784  | 0.122          | ∞%    | Breakfast    | 347 | -0.058         | ∞%   |
| Lunch        | 652  | -0.422         | ∞%    | Lunch        | 693 | -0.074         | ∞%   |
| Dinner       | 1001 | 0.396          | 51.4% | Dinner       | 381 | 0.057          | ∞%   |

| Rev.center 3 | MAE  | R <sup>2</sup> | MAPE  | Rev.center 4 | MAE | R <sup>2</sup> | MAPE |
|--------------|------|----------------|-------|--------------|-----|----------------|------|
| Breakfast    | 272  | -0.093         | ∞%    | Breakfast    |     |                |      |
| Lunch        | 1052 | 0.008          | 92.1% | Lunch        |     |                |      |
| Dinner       | 4812 | 0.317          | 43.8% | Dinner       | 0   | 0              | ∞%   |

| Rev.center 5 | MAE  | R <sup>2</sup> | MAPE  | Rev.center 6 | MAE  | R <sup>2</sup> | MAPE |
|--------------|------|----------------|-------|--------------|------|----------------|------|
| Breakfast    | 2227 | 0.701          | 15%   | Breakfast    | 219  | 0.063          | ∞%   |
| Lunch        | 4080 | -1.711         | ∞%    | Lunch        | 504  | 0.019          | ∞%   |
| Dinner       | 7678 | -0.149         | 66.8% | Dinner       | 1533 | 0.003          | ∞%   |

| Rev.center 7 | MAE | R <sup>2</sup> | MAPE | Rev.center 8 | MAE   | R <sup>2</sup> | MAPE |
|--------------|-----|----------------|------|--------------|-------|----------------|------|
| Breakfast    | 34  | -0.223         | ∞%   | Breakfast    | 1740  | 0.039          | ∞%   |
| Lunch        | 43  | -0.007         | ∞%   | Lunch        | 9879  | -0.074         | ∞%   |
| Dinner       | 87  | 0.022          | ∞%   | Dinner       | 17824 | 0.026          | ∞%   |

| Rev.center 9 | MAE | R <sup>2</sup> | MAPE |
|--------------|-----|----------------|------|
| Breakfast    | 221 | 0.053          | ∞%   |
| Lunch        | 164 | -0.09          | ∞%   |
| Dinner       | 87  | -0.104         | ∞%   |

- conclusion -

MAE(test) should be << 800 Rev center 2, 7, 9.

not suitable for Rev center 4 (Similar data), 8, 6

suitable for some meal periods in other revenue center

# Training data

| Rev. center 1 | MAE | R <sup>2</sup> | MAPE % |
|---------------|-----|----------------|--------|
| Breakfast     | 438 | 0.408          | ∞%     |
| Lunch         | 358 | 0.360          | ∞%     |
| Dinner        | 803 | 0.527          | 37.7%  |

| Rev. center 2 | MAE | R <sup>2</sup> | MAPE |
|---------------|-----|----------------|------|
| Breakfast     | 220 | 0.14           | ∞%   |
| Lunch         | 259 | 0.126          | ∞%   |
| Dinner        | 335 | 0.169          | ∞%   |

| Rev. center 3 | MAE  | R <sup>2</sup> | MAPE  |
|---------------|------|----------------|-------|
| Breakfast     | 259  | 0.031          | ∞%    |
| Lunch         | 737  | 0.172          | 64.9% |
| Dinner        | 2405 | 0.676          | 24.5% |

| Rev. center 4 | MAE | R <sup>2</sup> | MAPE |
|---------------|-----|----------------|------|
| Breakfast     |     |                |      |
| Lunch         |     |                |      |
| Dinner        | 0   | -0.001         | ∞%   |

| Rev. center 5 | MAE  | R <sup>2</sup> | MAPE   |
|---------------|------|----------------|--------|
| Breakfast     | 1529 | 0.839          | 9.6%   |
| Lunch         | 1585 | 0.924          | ∞%     |
| Dinner        | 4220 | 0.283          | 118.3% |

| Rev. center 6 | MAE | R <sup>2</sup> | MAPE |
|---------------|-----|----------------|------|
| Breakfast     | 152 | 0.131          | ∞%   |
| Lunch         | 274 | 0.219          | ∞%   |
| Dinner        | 766 | 0.003          | ∞%   |

| Rev. center 7 | MAE | R <sup>2</sup> | MAPE |
|---------------|-----|----------------|------|
| Breakfast     | 40  | 0.053          | ∞%   |
| Lunch         | 31  | 0.095          | ∞%   |
| Dinner        | 48  | 0.165          | ∞%   |

| Rev. center 8 | MAE   | R <sup>2</sup> | MAPE |
|---------------|-------|----------------|------|
| Breakfast     | 1072  | 0.072          | ∞%   |
| Lunch         | 757   | 0.221          | ∞%   |
| Dinner        | 12110 | 0.157          | ∞%   |

| Rev. center 9 | MAE | R <sup>2</sup> | MAPE |
|---------------|-----|----------------|------|
| Breakfast     | 141 | 0.313          | ∞%   |
| Lunch         | 152 | 0.114          | ∞%   |
| Dinner        | 66  | 0.116          | ∞%   |

\* This model is very suitable for rev. center 2, 7, 9  
rev. center 6, 7 (small and relatively equal)



target range 0-800

LSTM - 128  
epoch 200  
batch size 64

## Test\_data

| Rev. center 1 | MAE                     | R <sup>2</sup>              | MAPE                       | Rev. center 2 | MAE                       | R <sup>2</sup>             | MAPE                  |
|---------------|-------------------------|-----------------------------|----------------------------|---------------|---------------------------|----------------------------|-----------------------|
| Breakfast     | <del>760</del><br>714   | <del>0.16</del><br>0.126    | <del>00%</del><br>00%      | Breakfast     | <del>338</del><br>331     | <del>-0.036</del><br>0.009 | <del>00%</del><br>00% |
| Lunch         | <del>758</del><br>520   | <del>-0.963</del><br>0.148  | <del>00%</del><br>00%      | Lunch         | <del>481</del><br>532     | <del>0.01</del><br>0.049   | <del>00%</del><br>00% |
| Dinner        | <del>1158</del><br>1094 | <del>0.388</del><br>0.297   | <del>61.4%</del><br>58.4%  | Dinner        | <del>515</del><br>486     | <del>-0.008</del><br>-0.06 | <del>00%</del><br>00% |
| Rev. center 3 | MAE                     | R <sup>2</sup>              | MAPE                       | Rev. center 4 | MAE                       | R <sup>2</sup>             | MAPE                  |
| Breakfast     | <del>851</del><br>816   | <del>-0.171</del><br>-0.173 | <del>00%</del><br>00%      | Breakfast     |                           |                            |                       |
| Lunch         | <del>1045</del><br>1067 | <del>-2.491</del><br>0.02   | <del>155.6%</del><br>88.1% | Lunch         |                           |                            |                       |
| Dinner        | <del>5202</del><br>5024 | <del>0.274</del><br>0.288   | <del>49.7%</del><br>51.6%  | Dinner        | 0                         | 0                          | 00%                   |
| Rev. center 5 | MAE                     | R <sup>2</sup>              | MAPE                       | Rev. center 6 | MAE                       | R <sup>2</sup>             | MAPE                  |
| Breakfast     | <del>2180</del><br>4783 | <del>0.718</del><br>-0.238  | <del>14.1%</del><br>34.4%  | Breakfast     | <del>339</del><br>360     | <del>-0.069</del><br>-0.0  | <del>00%</del><br>00% |
| Lunch         | <del>2619</del><br>2628 | <del>-0.958</del><br>-0.242 | <del>00%</del><br>00%      | Lunch         | <del>468</del><br>570     | <del>0.1</del><br>-0.033   | <del>00%</del><br>00% |
| Dinner        | <del>5965</del><br>7203 | <del>-0.393</del><br>0.34   | <del>80.8%</del><br>110.1% | Dinner        | <del>1436</del><br>1762   | <del>0.04</del><br>0.004   | <del>00%</del><br>00% |
| Rev. center 7 | MAE                     | R <sup>2</sup>              | MAPE                       | Rev. center 8 | MAE                       | R <sup>2</sup>             | MAPE                  |
| Breakfast     | <del>38</del><br>33     | <del>-0.818</del><br>-0.191 | <del>00%</del><br>00%      | Breakfast     | <del>1673</del><br>1591   | <del>0.098</del><br>-0.046 | <del>00%</del><br>00% |
| Lunch         | <del>56</del><br>45     | <del>-0.738</del><br>-0.029 | <del>00%</del><br>00%      | Lunch         | <del>10013</del><br>9260  | <del>-0.067</del><br>0.001 | <del>00%</del><br>00% |
| Dinner        | <del>142</del><br>93    | <del>-1.127</del><br>-0.045 | <del>00%</del><br>00%      | Dinner        | <del>17758</del><br>20848 | <del>0.028</del><br>-0.098 | <del>00%</del><br>00% |
| Rev. center 9 | MAE                     | R <sup>2</sup>              | MAPE                       |               |                           |                            |                       |
| Breakfast     | <del>319</del><br>312   | <del>-0.775</del><br>-0.094 | <del>00%</del><br>00%      |               |                           |                            |                       |
| Lunch         | <del>149</del><br>158   | <del>-0.58</del><br>-0.009  | <del>00%</del><br>00%      |               |                           |                            |                       |
| Dinner        | <del>97</del><br>80     | <del>-0.168</del><br>-0.013 | <del>00%</del><br>00%      |               |                           |                            |                       |

suitable for: 2, 7, 9

# training data

| Rev. center 1 | MAE                   | R <sup>2</sup>            | MAPE                    |
|---------------|-----------------------|---------------------------|-------------------------|
| Breakfast     | <del>437</del><br>394 | <del>0.409</del><br>0.3   | <del>100%</del><br>100% |
| Lunch         | <del>351</del><br>420 | <del>0.418</del><br>0.205 | <del>100%</del><br>100% |
| Dinner        | <del>791</del><br>880 | <del>0.533</del><br>0.357 | <del>100%</del><br>100% |

| Rev. center 2 | MAE                   | R <sup>2</sup>            | MAPE                    |
|---------------|-----------------------|---------------------------|-------------------------|
| Breakfast     | <del>316</del><br>298 | <del>0.123</del><br>0.065 | <del>100%</del><br>100% |
| Lunch         | <del>316</del><br>284 | <del>0.088</del><br>0.068 | <del>100%</del><br>100% |
| Dinner        | <del>762</del><br>415 | <del>0.16</del><br>0.021  | <del>100%</del><br>100% |

| Rev. center 3 | MAE                     | R <sup>2</sup>             | MAPE                    |
|---------------|-------------------------|----------------------------|-------------------------|
| Breakfast     | <del>260</del><br>277   | <del>0.026</del><br>-0.013 | <del>100%</del><br>100% |
| Lunch         | <del>789</del><br>890   | <del>0.215</del><br>0.005  | <del>100%</del><br>100% |
| Dinner        | <del>3184</del><br>3240 | <del>0.612</del><br>0.57   | <del>100%</del><br>100% |

| Rev. center 4 | MAE               | R <sup>2</sup>    | MAPE                    |
|---------------|-------------------|-------------------|-------------------------|
| Breakfast     |                   |                   |                         |
| Lunch         |                   |                   |                         |
| Dinner        | <del>0</del><br>0 | <del>0</del><br>0 | <del>100%</del><br>100% |

| Rev. center 5 | MAE                     | R <sup>2</sup>            | MAPE                    |
|---------------|-------------------------|---------------------------|-------------------------|
| Breakfast     | <del>1861</del><br>2621 | <del>0.252</del><br>-0.64 | <del>100%</del><br>100% |
| Lunch         | <del>1169</del><br>1710 | <del>0.504</del><br>0.108 | <del>100%</del><br>100% |
| Dinner        | <del>4048</del><br>5557 | <del>0.346</del><br>0.066 | <del>100%</del><br>100% |

| Rev. center 6 | MAE                   | R <sup>2</sup>             | MAPE                    |
|---------------|-----------------------|----------------------------|-------------------------|
| Breakfast     | <del>124</del><br>162 | <del>0.125</del><br>-0.017 | <del>100%</del><br>100% |
| Lunch         | <del>269</del><br>319 | <del>0.215</del><br>0.095  | <del>100%</del><br>100% |
| Dinner        | <del>731</del><br>883 | <del>0.03</del><br>-0.007  | <del>100%</del><br>100% |

| Rev. center 7 | MAE                 | R <sup>2</sup>             | MAPE                    |
|---------------|---------------------|----------------------------|-------------------------|
| Breakfast     | <del>38</del><br>41 | <del>0.102</del><br>0.016  | <del>100%</del><br>100% |
| Lunch         | <del>89</del><br>95 | <del>0.858</del><br>-0.010 | <del>100%</del><br>100% |
| Dinner        | <del>46</del><br>53 | <del>0.919</del><br>0.076  | <del>100%</del><br>100% |

| Rev. center 8 | MAE                      | R <sup>2</sup>             | MAPE                    |
|---------------|--------------------------|----------------------------|-------------------------|
| Breakfast     | <del>780</del><br>756    | <del>0.067</del><br>-0.001 | <del>100%</del><br>100% |
| Lunch         | <del>725</del><br>6429   | <del>0.898</del><br>0.069  | <del>100%</del><br>100% |
| Dinner        | <del>1269</del><br>14896 | <del>0.169</del><br>-0.018 | <del>100%</del><br>100% |

| Rev. center 9 | MAE                   | R <sup>2</sup>             | MAPE                    |
|---------------|-----------------------|----------------------------|-------------------------|
| Breakfast     | <del>135</del><br>157 | <del>0.413</del><br>-0.231 | <del>100%</del><br>100% |
| Lunch         | <del>123</del><br>155 | <del>0.222</del><br>0.064  | <del>100%</del><br>100% |
| Dinner        | <del>70</del><br>72   | <del>0.167</del><br>-0.027 | <del>100%</del><br>100% |

suitable, 2, 7, 9.

target range 0-800

LSTM - 128

epoch 200

batch Size 64

## Test\_data

| Rev. center 1 | MAE                     | R <sup>2</sup>            | MAPE                       | Rev. center 2 | MAE                       | R <sup>2</sup>            | MAPE                |
|---------------|-------------------------|---------------------------|----------------------------|---------------|---------------------------|---------------------------|---------------------|
| Breakfast     | <del>649</del><br>748   | <del>0.198</del><br>0.138 | <del>∞%</del><br>∞%        | Breakfast     | <del>954</del><br>340     | <del>0.063</del><br>0.001 | <del>∞%</del><br>∞% |
| Lunch         | <del>522</del><br>618   | <del>0.022</del><br>0.135 | <del>∞%</del><br>∞%        | Lunch         | <del>476</del><br>533     | <del>0.035</del><br>0.024 | <del>∞%</del><br>∞% |
| Dinner        | <del>1021</del><br>1285 | <del>0.042</del><br>0.154 | <del>50.1%</del><br>64.2%  | Dinner        | <del>488</del><br>507     | <del>0.052</del><br>0.103 | <del>∞%</del><br>∞% |
| Rev. center 3 | MAE                     | R <sup>2</sup>            | MAPE                       | Rev. center 4 | MAE                       | R <sup>2</sup>            | MAPE                |
| Breakfast     | <del>285</del><br>280   | <del>0.022</del><br>0.016 | <del>∞%</del><br>∞%        | Breakfast     |                           |                           |                     |
| Lunch         | <del>1012</del><br>942  | <del>0.065</del><br>0.008 | <del>26.7%</del><br>30.1%  | Lunch         |                           |                           |                     |
| Dinner        | <del>4221</del><br>7129 | <del>0.231</del><br>0.029 | <del>46.1%</del><br>44.9%  | Dinner        | <del>0</del><br>0         | <del>0</del><br>0         | <del>∞%</del><br>∞% |
| Rev. center 5 | MAE                     | R <sup>2</sup>            | MAPE                       | Rev. center 6 | MAE                       | R <sup>2</sup>            | MAPE                |
| Breakfast     | <del>2190</del><br>3747 | <del>0.739</del><br>0.273 | <del>14%</del><br>26.5%    | Breakfast     | <del>230</del><br>267     | <del>0.054</del><br>0.015 | <del>∞%</del><br>∞% |
| Lunch         | <del>2493</del><br>2052 | <del>0.222</del><br>0.002 | <del>∞%</del><br>∞%        | Lunch         | <del>471</del><br>479     | <del>0.113</del><br>0.047 | <del>∞%</del><br>∞% |
| Dinner        | <del>2110</del><br>7567 | <del>0.219</del><br>0.16  | <del>40.3%</del><br>112.3% | Dinner        | <del>1805</del><br>1122   | <del>0.002</del><br>0.013 | <del>∞%</del><br>∞% |
| Rev. center 7 | MAE                     | R <sup>2</sup>            | MAPE                       | Rev. center 8 | MAE                       | R <sup>2</sup>            | MAPE                |
| Breakfast     | <del>35</del><br>31     | <del>0.965</del><br>0.101 | <del>∞%</del><br>∞%        | Breakfast     | <del>1908</del><br>1462   | <del>0.015</del><br>0.061 | <del>∞%</del><br>∞% |
| Lunch         | <del>59</del><br>42     | <del>0.912</del><br>0.003 | <del>∞%</del><br>∞%        | Lunch         | <del>16539</del><br>8882  | <del>1.422</del><br>0.062 | <del>∞%</del><br>∞% |
| Dinner        | <del>79</del><br>79     | <del>0.979</del><br>0.16  | <del>∞%</del><br>∞%        | Dinner        | <del>17730</del><br>20099 | <del>0.016</del><br>0.007 | <del>∞%</del><br>∞% |
| Rev. center 9 | MAE                     | R <sup>2</sup>            | MAPE                       |               |                           |                           |                     |
| Breakfast     | <del>313</del><br>222   | <del>1.006</del><br>0.001 | <del>∞%</del><br>∞%        |               |                           |                           |                     |
| Lunch         | <del>156</del><br>111   | <del>0.732</del><br>0.002 | <del>∞%</del><br>∞%        |               |                           |                           |                     |
| Dinner        | <del>92</del><br>79     | <del>0.932</del><br>0.001 | <del>∞%</del><br>∞%        |               |                           |                           |                     |

suitable for :-

# training data

| Rev. center 1 | MAE                   | R <sup>2</sup>          | MAPE                        | Rev. center 2 | MAE                 | R <sup>2</sup>         | MAPE                        |
|---------------|-----------------------|-------------------------|-----------------------------|---------------|---------------------|------------------------|-----------------------------|
| Breakfast     | $\frac{36}{41}$       | $\frac{0.102}{0.016}$   | $\frac{\infty\%}{\infty\%}$ | Breakfast     | $\frac{475}{449}$   | $\frac{0.113}{0.047}$  | $\frac{\infty\%}{\infty\%}$ |
| Lunch         | $\frac{89}{95}$       | $\frac{0.352}{0.010}$   | $\frac{\infty\%}{\infty\%}$ | Lunch         | $\frac{1405}{1172}$ | $\frac{-0.002}{0.013}$ | $\frac{\infty\%}{\infty\%}$ |
| Dinner        | $\frac{155}{157}$     | $\frac{0.412}{0.231}$   | $\frac{\infty\%}{\infty\%}$ | Dinner        | $\frac{2031}{1161}$ | $\frac{-0.06}{0.02}$   | $\frac{\infty\%}{\infty\%}$ |
| Rev. center 3 | MAE                   | R <sup>2</sup>          | MAPE                        | Rev. center 4 | MAE                 | R <sup>2</sup>         | MAPE                        |
| Breakfast     | $\frac{149}{158}$     | $\frac{-0.259}{-0.009}$ | $\frac{\infty\%}{\infty\%}$ | Breakfast     |                     |                        |                             |
| Lunch         | $\frac{97}{80}$       | $\frac{-0.168}{0.013}$  | $\frac{\infty\%}{\infty\%}$ | Lunch         |                     |                        |                             |
| Dinner        | $\frac{224}{201}$     | $\frac{0.22}{-0.05}$    | $\frac{\infty\%}{\infty\%}$ | Dinner        | $\frac{0}{0}$       | $\frac{0}{0}$          | $\frac{\infty\%}{\infty\%}$ |
| Rev. center 5 | MAE                   | R <sup>2</sup>          | MAPE                        | Rev. center 6 | MAE                 | R <sup>2</sup>         | MAPE                        |
| Breakfast     | $\frac{7253}{6439}$   | $\frac{0.348}{0.049}$   | $\frac{\infty\%}{\infty\%}$ | Breakfast     | $\frac{381}{338}$   | $\frac{0.32}{-0.016}$  | $\frac{\infty\%}{\infty\%}$ |
| Lunch         | $\frac{1861}{14596}$  | $\frac{0.169}{-0.018}$  | $\frac{\infty\%}{\infty\%}$ | Lunch         | $\frac{1017}{942}$  | $\frac{-0.05}{0.208}$  | $\frac{76.7\%}{\infty\%}$   |
| Dinner        | $\frac{10153}{13062}$ | $\frac{0.02}{-0.05}$    | $\frac{\infty\%}{\infty\%}$ | Dinner        | $\frac{493}{713}$   | $\frac{0.275}{0.029}$  | $\frac{46.1\%}{\infty\%}$   |
| Rev. center 7 | MAE                   | R <sup>2</sup>          | MAPE                        | Rev. center 8 | MAE                 | R <sup>2</sup>         | MAPE                        |
| Breakfast     | $\frac{38}{26}$       | $\frac{0.01}{0.01}$     | $\frac{\infty\%}{\infty\%}$ | Breakfast     | $\frac{38}{41}$     | $\frac{0.102}{0.016}$  | $\frac{\infty\%}{\infty\%}$ |
| Lunch         | $\frac{51}{39}$       | $\frac{-0.16}{-0.015}$  | $\frac{\infty\%}{\infty\%}$ | Lunch         | $\frac{89}{95}$     | $\frac{0.352}{-0.010}$ | $\frac{\infty\%}{\infty\%}$ |
| Dinner        | $\frac{241}{121}$     | $\frac{0.53}{-0.13}$    | $\frac{\infty\%}{\infty\%}$ | Dinner        | $\frac{61}{75}$     | $\frac{0.01}{-0.25}$   | $\frac{\infty\%}{\infty\%}$ |
| Rev. center 9 | MAE                   | R <sup>2</sup>          | MAPE                        |               |                     |                        |                             |
| Breakfast     | $\frac{39}{33}$       | $\frac{-0.214}{-0.191}$ | $\frac{\infty\%}{\infty\%}$ |               |                     |                        |                             |
| Lunch         | $\frac{56}{45}$       | $\frac{-0.238}{-0.029}$ | $\frac{\infty\%}{\infty\%}$ |               |                     |                        |                             |
| Dinner        | $\frac{142}{73}$      | $\frac{-1.123}{-0.045}$ | $\frac{\infty\%}{\infty\%}$ |               |                     |                        |                             |

This method is very

## Final conclusion

suitable for 2, 7, 9.